

KENYA INSTITUTE OF PRIMATE RESEARCH

Ministry of Health

Institutional Data Governance Development Plan

2026–2027

This Development Plan provides the strategic road-map for the progressive development and implementation of the KIPRE IDGF in alignment with the KIPRE Strategic Plan (2023–2027).

Document control

Document information	
Document title	Institutional Data Governance Framework Development Plan
Document number	KIPRE/DSAS/DGF/PLN/001
Document owner	Data Science and Analytics Section (DSAS)
Prepared by	Mr. Patrick W. Mwaura- Lead DSAS and DPO
Reviewed by	Director, Capacity Building, Partnership and Grant Management
Signature and Date	
Approved by	Director General
Signature and Date	
Effective date	
Next review date	
Distribution	Internal
Document classification	Internal Use Only

Revision history

Version	Date	Description of revision	Responsible officer
1.0	31 July 2026	Initial issue.	Mr. Patrick W. Mwaura

Executive summary

The KIPRE Strategic Plan (2023–2027) identifies the establishment of an Institutional Data Governance Framework (IDGF) as a strategic institutional initiative for strengthening the governance, management and stewardship of institutional data assets. This Development Plan provides the implementation roadmap for progressively establishing the Framework and operationalizing this strategic objective across the Institute.

Building upon the approved Data Management and Protection Policy, the Development Plan adopts a structured and phased approach for developing the governance structures, governance policies, Standard Operating Procedures (SOPs), standards, guidelines and supporting implementation instruments required to establish a comprehensive institutional data governance program.

Implementation of the IDGF will be undertaken through three progressive phases:

- **Phase I: Framework operationalization** focuses on operationalizing the approved Data Management and Protection Policy through the establishment of governance structures, implementation mechanisms, a network of Data Governance Champions, institutional capacity development and the development of priority Standard Operating Procedures.
- **Phase II: Framework expansion** focuses on strengthening the governance framework through the progressive development of additional governance policies, standards, guidelines and other supporting governance instruments required to support institutional data governance.
- **Phase III: Framework institutionalization** focuses on organisation-wide implementation of the completed Framework through capacity building, monitoring, evaluation and continual improvement to ensure its long-term sustainability.

Successful implementation of this Development Plan will establish an integrated institution-wide data governance framework that strengthens institutional accountability, promotes responsible stewardship of institutional data assets, enhances data quality, supports regulatory compliance, facilitates responsible research and innovation, and improves evidence-informed decision-making in advancing the Institute's mandate.

Contents

List of abbreviations	5
1 Background context	6
2 Introduction to the Plan	7
2.1 Plan purpose and objectives	7
2.2 Plan Structure	8
 PART I	
 Status of institutional data governance landscape	9
2.3 Current state of institutional data governance	9
2.4 Existing institutional governance structures	10
2.4.1 Board of Directors	10
2.4.2 Director General	10
2.4.3 ISREC	11
2.4.4 Directorate of Research and Product Development (DRPD)	11
2.4.5 DSAS	11
2.4.6 DPO Office	11
2.4.7 ICT Division	11
2.5 Proposed institutional data governance architecture	12
2.6 Proposed institutional governance document suite	12
2.6.1 Governance policies	15
 PART II	
 Development road-map	17
2.7 Development approach	17
2.8 Guiding principles	17
2.9 Governance maturity pathway	19
2.10 Critical success factors	19
2.11 Phased development strategy	19
2.12 Phase I: Operationalizing the Data Management and Protection Policy	21
2.12.1 Objectives	21
2.12.2 Data protection compliance implementation strategy	21
2.12.3 SOPs inventory	22
2.12.4 Priority SOPs for the DPO Office	26
2.12.5 Supporting implementation documents	27
2.12.6 Expected outputs	28
2.13 Phase II: Expansion of the governance framework	29
2.13.1 Purpose of Phase II	29
2.13.2 Governance policies to be developed	29
2.13.3 Supporting governance instruments	29

2.13.4 Governance structures and stewardship	30
2.14 Phase III: Institutionalization of the IDGF	31
2.14.1 Framework integration	31
2.14.2 Capacity building and cultural sensitization	31
2.14.3 Monitoring, evaluation, and compliance assurance	31
2.14.4 Continuous improvement and evolutionary governance	32
3 Overall implementation road-map	32
4 Conclusion	32
5 References	34
References	34

List of Tables

1 Institutional data governance document hierarchy.	14
2 Governance policy portfolio proposed under the Institutional Data Governance Framework.	15
3 Proposed SOPs required to operationalize the Data Management and Protection Policy.	25
4 Priority SOPs for the DPO Office.	26
5 Proposed supporting implementation documents for the Phase I priority Standard Operating Procedures.	28
6 Proposed governance policies and framework additions for development under Phase II.	29
7 Proposed governance structure and stewardship responsibilities under the IDGF.	30
8 Overall implementation road-map for the IDGF.	33

List of Figures

1 Proposed architecture of the Institutional Data Governance Framework showing the governance hierarchy from institutional governance structures through governance instruments and implementation mechanisms.	13
--	----

List of abbreviations

DPA	Data Protection Act, 2019
GDPR	Data Protection (General) Regulations, 2021
AI	Artificial Intelligence
DGF	Data Governance Framework
DMP	Data Management Plan
DPO	Data Protection Officer
DSAS	Data Science and Analytics Section
ICT	Information and Communication Technology
IDGF	Institutional DGF
ISREC	Institutional Scientific and Research Ethics Committee
KIPRE	Kenya Institute of Primate Research
ODPC	Office of the Data Protection Commissioner
PI	Principal Investigator
RDM	Research Data Management
SOP	Standard Operating Procedure

1 Background context

Institutional data have evolved into a vital strategic asset that underpins cutting-edge research, technological innovation, enterprise management, and evidence-informed decision-making. The rapid proliferation of artificial intelligence (AI), cloud computing architectures, advanced analytics, and distributed digital nodes has exponentially escalated both the volume and systemic complexity of data generated within research environments. Consequently, data governance has transitioned from a localized information technology function into a critical corporate and fiduciary imperative. This evolution demands comprehensive frameworks encompassing legal compliance, absolute accountability, rigorous stewardship, and ethical exploitation across the entire data lifecycle.

On the global stage, premier scientific institutions are systematically adopting integrated data governance frameworks to anchor consistency, security, and cross-border operational resilience. For biomedical and preclinical research centers, high-fidelity datasets are the bedrock of scientific integrity, translational drug discovery, predictive machine learning models, and the formulation of reliable evidence for public health policy. Effective governance establishes the structural and administrative mechanisms necessary to guarantee that high-value intellectual assets remain secure, structurally accurate, seamless in interoperability, and functionally fit for purpose.

Within the Kenyan landscape, data administration is dictated by a strict and rapidly evolving statutory architecture that aggressively champions digital transformation while safeguarding informational privacy. Key legislative instruments underpinning this domain include the Constitution of Kenya, 2010; the Data Protection Act, 2019; the Computer Misuse and Cybercrimes Act, 2018; the Access to Information Act, 2016; the Kenya Information and Communications Act; the Public Archives and Documentation Service Act; and the Science, Technology and Innovation Act, 2013. Collectively, these regulatory instruments legally obligate public state corporations to institute formalized internal mechanisms that ensure institutional data are processed lawfully, transparently, and securely throughout their operational existence.

As Kenya's premier semi-autonomous government agency for biomedical research under the Ministry of Health, KIPRE generates and curates highly specialized streams of scientific, preclinical laboratory, epidemiological surveillance, and corporate administrative records. These data assets directly drive national health priorities, Universal Health Coverage (UHC) initiatives, and strategic international partnerships. Comprehensive data governance is therefore paramount to shielding proprietary intellectual property, maintaining strict regulatory compliance, eliminating insider fraud vulnerabilities, and reinforcing public and donor trust in the Institute's operations.

Recognizing these strategic mandates, the KIPRE Strategic Plan (2023–2027) explicitly identifies the formulation of an Institutional Data Governance Framework (IDGF) as a landmark corporate objective. This roadmap was directly informed by an exhaustive, organization-wide Data Management Needs Assessment spearheaded by the Data Science and Analytics Section (DSAS). The diagnostic assessment exposed critical gaps, highlighting an immediate institutional need for harmonized governance policies, explicitly defined data ownership structures, cross-functional standard operating procedures, technical validation standards, and enhanced internal accountability matrices.

KIPRE has since secured an important compliance milestone by enacting its primary Data Management and Protection Policy and achieving dual registration with the Office of the Data Protection Commissioner (ODPC) as both a Data Controller and a Data Processor. While these achievements

satisfy immediate legal prerequisites, they represent only the baseline threshold of an enterprise-wide data governance ecosystem. True operational maturity requires a broader supporting document suite—comprising targeted technical guidelines, localized standard operating procedures, controlled data capture forms, and immutable internal audit logs—that systematically hardcodes data minimization and protection into daily laboratory and administrative tasks.

The progressive execution of the IDGF represents the next vital phase in KIPRE's data maturity trajectory. This Development Plan delivers the strategic, phased blueprint for transforming raw information assets into legally shielded and optimized institutional resources, empowering KIPRE to fulfill its scientific research mandate within a highly secure, digital, and data-intensive environment.

2 Introduction to the Plan

The Institutional Data Governance Framework (IDGF) serves as the definitive architecture through which KIPRE orchestrates a coordinated strategy for the management, protection, stewardship, and utilization of data throughout its lifecycle. This Framework establishes the vital organizational structures, governance instruments, and accountability mechanisms necessary to ensure that institutional data assets are handled consistently, securely, and in strict alignment with legal, statutory, and regulatory benchmarks.

The formal endorsement of the Data Management and Protection Policy marked the inaugural milestone of this architecture, laying the structural foundation for robust data oversight. However, advancing toward a mature governance posture requires the incremental evolution and implementation of complementary policies, technical standards, Standard Operating Procedures (SOPs), and operational guidelines that collectively secure and optimize KIPRE's data assets.

This Development Plan outlines the strategic roadmap for the progressive operationalization of the IDGF. It identifies the critical governance components required for immediate deployment, defines the systemic arrangements necessary to sustain the framework, assigns departmental execution responsibilities, and presents a phased implementation timeline across all tiers of the Institute.

2.1 Plan purpose and objectives

The primary purpose of this Development Plan is to provide an operational, step-by-step roadmap for the progressive integration of the Institutional Data Governance Framework at KIPRE.

Specifically, the Plan aims to:

1. Operationalize the approved Data Management and Protection Policy through a structured, enterprise-wide compliance program;
2. Establish robust governance bodies, institutional mandates, and enforcement mechanisms to sustain long-term data integrity;
3. Provide a clear, phased methodology for drafting technical standards, cross-functional SOPs, and tactical implementation templates;
4. Strengthen internal human resource capacity for risk mitigation, responsible data handling, and statutory data protection compliance; and

5. Institutionalize an integrated, organization-wide data governance ecosystem that underpins scientific research excellence, technological innovation, and evidence-based clinical decision-making.

2.2 Plan Structure

This Development Plan is partitioned into two complementary, strategic segments that collectively drive the establishment of the IDGF:

- **Part I: Institutional Landscape** delineates the current data governance environment, proposes the target data architecture, and maps out the required suite of controlled governance instruments.
- **Part II: Development Road-map** establishes the operational roll-out strategy, defines the core guiding principles, and presents the phased methodology through which the framework will be executed, monitored, and institutionalized across KIPRE.

Together, these sections provide a unified, coherent blueprint for transforming KIPRE's data assets into secure, compliant, and strategically leveraged institutional resources.

PART I

Status of institutional data governance landscape

2.3 Current state of institutional data governance

KIPRE has made significant progress towards strengthening institutional data governance over recent years.

This progress has been driven by increasing reliance on digital technologies to support research, administration, and institutional operations. The Institute has progressively adopted several integrated digital systems, including the Enterprise Resource Planning (ERP) system, Human Resource Information System (HRIS), biometric attendance management system, electronic procurement (e-GP), financial management networks, and Laboratory Information Management Systems (LIMS), alongside other specialized digital platforms supporting research and corporate functions.

These platforms continue to generate substantial volumes of operational, financial, and administrative data that require coordinated governance to ensure their long-term data quality, schema integrity, baseline security, and effective utilization across all organizational levels.

At the same time, the Institute has experienced continued growth in high-impact, multidisciplinary research programmes involving human participants, non-human primates, pre-clinical laboratory investigations, genomic sequencing, infectious disease surveillance, environmental health, and One Health initiatives executed in partnership with national and international collaborators. These activities generate diverse, highly sensitive categories of institutional information, including clinical research datasets, molecular laboratory records, surveillance data, biological specimen metadata, intellectual property disclosures, and corporate administrative assets.

The increasing volume, diversity, and strategic importance of these data require robust governance mechanisms that promote technical consistency, full accountability, and responsible stewardship throughout the data life-cycle, preventing unauthorized disclosure or loss of data availability.

Recognizing these emerging needs, the KIPRE Strategic Plan 2023–2027 identifies the enhancement of data management, advanced analytics, and data protection strategies as a core strategic institutional priority and specifically provides for the development and implementation of an integrated IDGF. This strategic commitment reflects the Institute's recognition that data are strategic institutional assets that support research excellence, operational efficiency, technology transfer, regulatory compliance, and evidence-informed decision-making.

In support of this strategic direction, the Institute has already established several critical governance foundations through a coordinated portfolio of master policies. These include the formal approval of the Data Management and Protection Policy (POL/KIPRE/DRPD/002, Version 01), which establishes baseline privacy rules; the KIPRE Research Policy (POL/KIPRE/DRPD/001, Version 01), which governs the scientific data lifecycle and mandates international FAIR data principles; the Intellectual Property Policy (POL/KIPRE/DCBPGM/001, Version 01), which regulates the disclosure, ownership, and commercialization parameters of institutional data assets; and the foundational

KIPRE ICT Policy, which mandates operational baseline technical controls and endpoint security configurations.

Furthermore, the Institute has achieved successful registration with the Office of the Data Protection Commissioner (ODPC) as both a Data Controller and a Data Processor, finalized the formal designation of the institutional DPO Office, and completed a comprehensive, organization-wide Data Management Needs Assessment to systematically map existing data handling practices, governance gaps, and future technical requirements.

Collectively, these milestones provide a strong compliance and operational foundation upon which a comprehensive IDGF can be progressively developed. However, while these achievements establish the Institute's initial governance environment, they do not yet constitute a fully operational institutional data governance framework.

Several governance components—including additional technical policies, Standard Operating Procedures (SOPs), data quality standards, operational guidelines, controlled intake forms, system implementation templates, centralized registers, and formal enforcement mechanisms—remain to be developed to ensure coordinated management of institutional data assets throughout their entire operational life-cycle.

The progressive development of these governance instruments therefore represents the next logical stage in strengthening institutional data governance, advancing KIPRE towards a mature, sustainable, and enterprise-wide data governance program.

2.4 Existing institutional governance structures

Institutional data governance at KIPRE is currently supported through a number of governance structures established to fulfill specific statutory, administrative, research and operational responsibilities.

Although these structures were established independently to support different institutional functions, each contributes directly or indirectly to the governance of institutional data. Collectively, they provide an important governance foundation upon which the IDGF will be progressively developed.

2.4.1 Board of Directors

The Board of Directors provides the highest level of institutional oversight and is ultimately responsible for ensuring that the Institute establishes appropriate governance, risk management, and accountability mechanisms. The Board approves master institutional policies and provides strategic direction for the governance of KIPRE's information assets, intellectual property parameters under POL/KIPRE/DCBPGM/001, and master research repositories.

2.4.2 Director General

The Director General is responsible for the overall implementation of institutional policies and provides executive oversight for data governance operations. In accordance with the statutory mandates of the Institute, the Office of the Director General ensures that appropriate governance resources, funding frameworks, and administrative arrangements are established to support the secure management of institutional data, granting final authorization for data sharing, Material Transfer

Agreements (MTAs), and specialized Data Use Agreements (DUAs).

2.4.3 ISREC

The Institutional Scientific and Research Ethics Committee (ISREC) provides oversight for the scientific and ethical conduct of research involving human participants, animals, and other regulated biological materials. The Committee operates through two specialized boards: the Animal Research Ethics Committee (AREC), which conducts comprehensive harm-benefit analyses of animal experimentation, and the Human Research Ethics Committee (HREC), which reviews the use of personal health records, genetic datasets, and tissue samples. ISREC serves as a critical governance interface ensuring that ethical clearance tracks run concurrently with data privacy vetting, as co-mandated by the KIPRE Research Policy (POL/KIPRE/DRPD/001).

2.4.4 Directorate of Research and Product Development (DRPD)

The DRPD coordinates institutional research programs and oversees project implementation, monitoring, and reporting metrics. The Directorate ensures that research datasets are generated, managed, and archived within centralized systems in accordance with approved institutional policies, preserving scientific integrity and ensuring adherence to program agreements and investigator stewardship rules.

2.4.5 DSAS

The Data Science and Analytics Section (DSAS) serves as the Institute's technical focal point for data management, advanced analytics, database integration, and data visualization. Under the KIPRE Strategic Plan 2023–2027, the Section is tasked with leading the design and development of the overarching IDGF and supporting institutional frameworks governing data stewardship, data quality standards (ISO 8000), and automated data pipelines.

2.4.6 DPO Office

The Data Protection Officer (DPO) Office provides institutional oversight for compliance with Kenya's Data Protection Act, 2019 and associated regulations. Mapped directly to the Deputy Director of the ICT Division and reporting independently to the Director General, the DPO Office is responsible for monitoring personal data processing, maintaining the centralized annual data risk register, managing data subject rights (DSAR) workflows, and coordinating statutory 72-hour breach notifications to the ODPC.

2.4.7 ICT Division

The ICT Division provides the underlying technological infrastructure supporting KIPRE's information systems and database networks in direct alignment with the active KIPRE ICT Policy. The Division is responsible for network administration, server security, user access privilege management, secure daily backup routines, and the implementation of robust technical, organizational, and physical safeguards necessary to maintain data availability, integrity, and confidentiality.

2.5 Proposed institutional data governance architecture

The IDGF is envisioned as the overarching governance architecture through which the Institute will coordinate the management, protection, quality, accessibility, sharing and utilization of institutional data assets. Rather than establishing an entirely new governance structure, the Framework will integrate existing governance mechanisms, strengthen institutional coordination and provide a coherent governance environment for managing institutional data throughout its life-cycle.

The Framework is intended to provide a common institutional approach to data governance by establishing clear governance principles, institutional responsibilities, decision-making structures and supporting governance instruments. It will facilitate harmonization of governance practices across research, laboratory, administrative and corporate functions while ensuring alignment with applicable legal, regulatory and institutional requirements.

The proposed IDGF consists of two complementary components:

- a) **Governance structures**, which provide institutional leadership, oversight, accountability and decision-making; and
- b) **Governance documents**, which provide the policies, procedures, standards and implementation tools required to operationalize governance across the Institute.

The governance structures described in the previous section will continue to provide institutional oversight within their respective mandates. The governance documents will progressively establish a structured hierarchy that translates institutional objectives into operational practice while supporting consistency, accountability and continual improvement.

The proposed governance architecture is illustrated in Figure 1.

The purpose of each governance document category is summarized in Table 1.

The governance document hierarchy provides the operational foundation upon which the IDGF will be progressively developed. Consistent with the phased implementation approach presented in this Development Plan, Phase I will focus on operationalizing the existing Data Management and Protection Policy through the development of supporting SOPs and their associated forms, templates, notices and institutional registers required for effective implementation. Subsequent phases will progressively expand the Framework through the development of additional governance policies and their corresponding implementation documents until a comprehensive, integrated and sustainable IDGF is established.

2.6 Proposed institutional governance document suite

The IDGF will ultimately comprise a coordinated suite of governance documents that collectively support the governance of institutional data assets throughout their life-cycle. While the Institute has already established important governance foundations, the Framework envisions the progressive development of additional governance instruments to address emerging institutional, scientific and regulatory requirements.

The proposed document suite is intended to guide future governance development and does not imply that all documents will be developed simultaneously. Rather, implementation will be undertaken in phases according to institutional priorities, available resources and strategic needs.

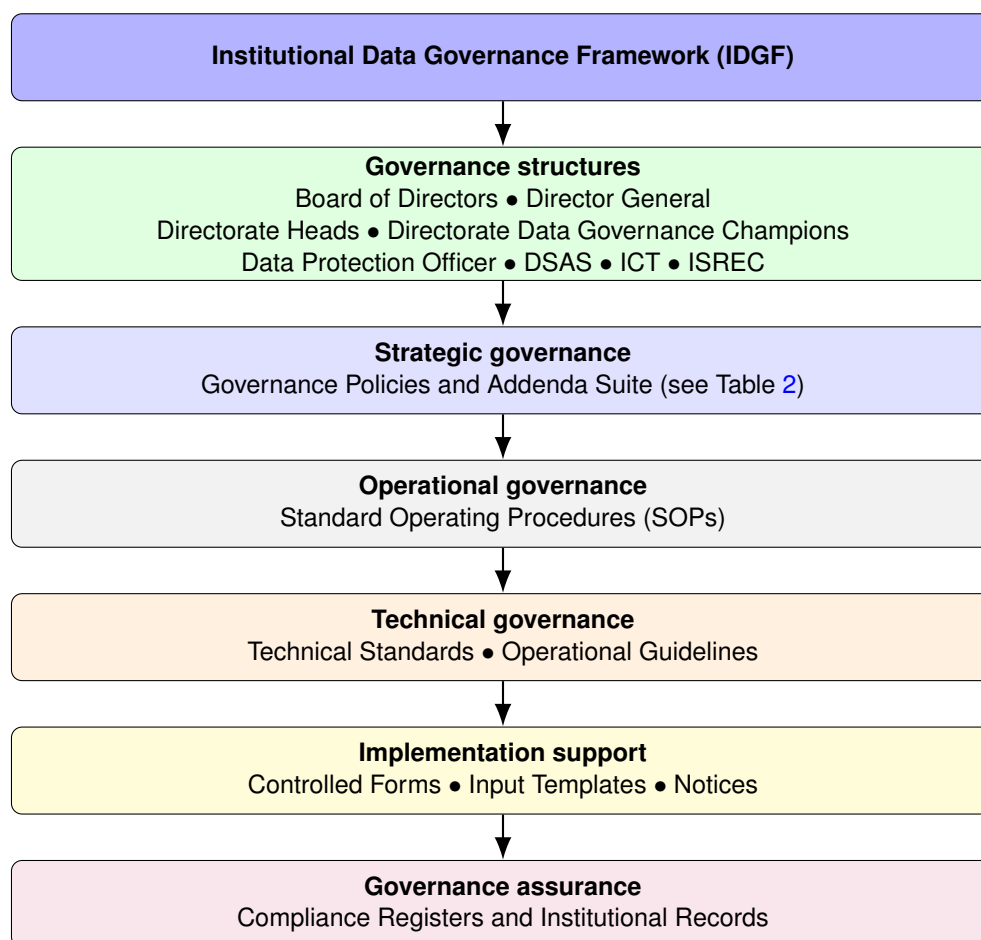


Figure 1: Proposed architecture of the Institutional Data Governance Framework showing the governance hierarchy from institutional governance structures through governance instruments and implementation mechanisms.

Governance layer	Governance instrument	Purpose
Strategic	Policies and Addenda	Establish institutional governance principles, strategic direction, and management commitment. Policies translate applicable legal, regulatory, and institutional requirements into institutional data governance visions.
Operational	Standard Operating Procedures (SOPs)	Define institutional workflows, cross-functional responsibilities, operational controls, and decision-making timelines required to implement approved policies consistently across all KIPRE directorates.
Technical	Standards and guidelines	Provide technical specifications, minimum security requirements, recommended practices, and baseline implementation guidance to promote consistency, data interoperability, and continual maturity.
Implementation	Forms, templates, notices and checklists	Provide controlled, audit-ready instruments that standardize data capture activities, support operational execution, and ensure consistent documentation tracks across institutional processes.
Assurance	Registers and institutional records	Provide immutable forensic evidence of implementation, facilitate monitoring, internal audit loops, corporate reporting, and continual improvement while demonstrating regulatory data compliance.

Table 1: Institutional data governance document hierarchy.

2.6.1 Governance policies

The proposed governance policy portfolio and its legal and institutional basis are presented in Table 2.

Governance area	Governance policy / Instrument	Legal and regulatory basis	Impl/Dev. phase	Purpose
Privacy and data protection	Data Management and Protection Policy	Constitution of Kenya (Art. 31); Data Protection Act, 2019; Data Protection Regulations, 2021 (LN. 263, 264, 265)	Existing	Provides the foundational institutional framework for the lawful collection, processing, storage, sharing, retention, and protection of personal and institutional data assets.
Artificial intelligence governance	Artificial Intelligence Governance Policy	Kenya National AI Strategy; Data Protection Act, 2019; UNESCO Recommendation on the Ethics of AI	Phase II	Establishes clear strategic principles for the responsible deployment, risk mitigation, algorithmic fairness, and oversight of AI and machine learning systems.
Data quality governance	Data Quality Standard Addendum (KIPRE ICT Policy Addendum)	KIPRE Strategic Plan 2023–2027; ISO 8000 (series); DAMA-DMBOK2	Phase II	Enforces technical database engineering standards for validating data accuracy, completeness, consistency, and structural integrity across the LIMS platform.
Research data lifecycle	Research Data Lifecycle Guidelines (KIPRE Research Policy Addendum)	Science, Technology and Innovation Act, 2013; NACOSTI research requirements; KIPRE Strategic Plan 2023–2027	Phase II	Provides structural guidelines for research data throughout the scientific lifecycle, optimizing dataset custody, long-term stewardship, and archiving protocols.
Open science and data sharing	Open Science and FAIR Data Sharing Guidelines	UNESCO Recommendation on Open Science; Data Protection Act, 2019; institutional and donor data-sharing requirements	Phase II	Promotes responsible data sharing, public open science practices, and multi-institutional collaborations while securing underlying intellectual property boundaries.

Table 2: Governance policy portfolio proposed under the Institutional Data Governance Framework.

The governance policies presented above collectively define the target policy portfolio for the proposed IDGF. While the Data Management and Protection Policy provides the Institute with an important governance foundation, the remaining policies and technical guidelines will be progressively developed to establish a comprehensive, integrated and sustainable governance framework capable of supporting the Institute's evolving research, operational and digital transformation agenda.

The development of these policies will be undertaken in phases in accordance with institutional priorities, available resources and strategic requirements. The proposed phased development approach is presented in Part II of this Development Plan.

PART II

Development road-map

2.7 Development approach

The development of the IDGF will be undertaken through a phased and incremental approach that recognizes the Institute's existing governance achievements while progressively expanding institutional data governance maturity.

This approach acknowledges that the establishment of a comprehensive governance framework is a continuous institutional process requiring coordinated planning, prioritization and sustained implementation rather than the development of isolated governance documents.

The proposed approach is founded on the principle of progressive governance maturity. Rather than attempting to develop all governance instruments simultaneously, the Institute will build upon its existing governance foundations by first strengthening and operationalising approved governance instruments before expanding into additional governance domains.

This strategy promotes efficient utilization of institutional resources, minimizes duplication of effort and ensures that each governance component is fully embedded before subsequent governance instruments are introduced.

The phased approach also recognizes that institutional data governance extends beyond regulatory compliance. While compliance with the DPA and related legal and regulatory requirements remains an important objective, the broader purpose of the Framework is to establish governance structures that enhance institutional decision-making, improve research quality, strengthen data stewardship, support digital transformation and maximize the strategic value of institutional data assets.

The implementation priorities presented in this Development Plan have been informed by the Institute's current governance landscape, the findings of the Institutional Data Management Needs Assessment, strategic priorities identified in the KIPRE Strategic Plan 2023–2027, and emerging national and international developments in data governance. Collectively, these provide a practical and evidence-based road-map for progressively strengthening institutional governance of data.

The development road-map is organized into three complementary phases. Phase I focuses on operationalising the existing Data Management and Protection Policy through the development of its supporting operational documentation. Phase II expands the governance portfolio through the development of additional governance policies and their supporting governance instruments. Phase III consolidates these governance instruments into a fully integrated IDGF supported by institutional capacity building, implementation, monitoring and continual improvement.

2.8 Guiding principles

The development and implementation of the IDGF shall be guided by principles that promote responsible, consistent and sustainable governance of institutional data assets. These principles provide the foundation upon which governance policies, standard operating procedures, standards,

guidelines and supporting governance instruments will be developed and implemented across the Institute.

The Framework shall be guided by the following principles:

1. Strategic alignment

The IDGF shall support the strategic objectives of KIPRE and remain aligned with the Institute's mandate, Strategic Plan, applicable legal and regulatory requirements, and emerging national priorities relating to digital transformation, data governance and innovation.

2. Institution-wide governance

Data governance shall be implemented as an institution-wide responsibility. All directorates, divisions, programs, projects, sections and governance structures involved in the generation, collection, processing, storage, sharing and use of institutional data shall contribute to the implementation of the Framework in accordance with their respective mandates and responsibilities.

3. Risk-based implementation

Governance measures shall be proportionate to the nature, sensitivity, volume and intended use of institutional data. Higher-risk activities shall be subject to enhanced governance, oversight and compliance requirements.

4. Life-cycle management

Institutional data shall be governed throughout their entire life-cycle, including planning, collection, creation, storage, use, sharing, archival and secure disposal, while ensuring appropriate quality, integrity, security and accountability at each stage.

5. Compliance by design

Governance instruments developed under the Framework shall incorporate applicable legal, regulatory and institutional requirements from the outset, thereby promoting compliance by design rather than corrective implementation.

6. Standardization and consistency

Institutional governance processes shall be supported through standardized policies, SOPs, standards, guidelines, forms, templates, notices and registers to ensure consistency, transparency and efficiency across the Institute.

7. Stewardship and accountability

Clear governance roles, responsibilities and decision-making authority shall be assigned for institutional data assets. Data owners, custodians, stewards and users shall remain accountable for their respective governance responsibilities.

8. Continual improvement

The IDGF shall evolve progressively through regular monitoring, stakeholder engagement, periodic review, lessons learned and emerging best practices to ensure that governance ar-

rangements remain effective, relevant and responsive to changing institutional and technological environments.

2.9 Governance maturity pathway

The implementation of the IDGF adopts a governance maturity approach that enables KIPRE to progressively strengthen its institutional data governance capabilities. Rather than attempting to establish the entire Framework simultaneously, implementation is structured into sequential phases that progressively build governance capacity, institutional ownership and organizational maturity.

Phase I establishes the foundational governance capability by operationalising the approved Data Management and Protection Policy and strengthening institutional data protection compliance. Phase II expands the governance framework through the development of additional governance policies and supporting governance instruments that collectively form the broader IDGF. Phase III focuses on institutionalizing the completed Framework through organization-wide implementation, capacity building, monitoring, evaluation and continual improvement.

This progressive implementation approach allows governance capabilities to mature in a coordinated and sustainable manner while ensuring that each phase provides the foundation for the successful implementation of the next.

2.10 Critical success factors

Successful implementation of the IDGF will depend upon sustained institutional commitment and coordinated participation across all organizational levels. The following factors will be critical to the successful implementation of this Development Plan:

- Strong leadership commitment and management support.
- Clear governance structures with defined roles, responsibilities and accountability.
- Active participation of Directorates, Divisions and Sections in implementing approved governance instruments.
- Establishment of a functional network of Data Protection Champions to coordinate institutional compliance activities.
- Adequate financial, technical and human resources to support phased implementation.
- Continuous staff awareness, capacity building and change management.
- Regular monitoring, evaluation and continual improvement of institutional governance practices.

2.11 Phased development strategy

The IDGF will be developed progressively through three implementation phases. Each phase builds upon the outputs of the preceding phase, allowing the Institute to establish governance capabilities incrementally while ensuring that immediate institutional priorities are addressed and available resources are utilized efficiently.

The phased approach recognizes that KIPRE has already established important governance foundations through the approval of the Data Management and Protection Policy, registration with the Office of the ODPC, establishment of the DPO Office, and the strategic commitment to develop an IDGF under the KIPRE Strategic Plan 2023–2027.

Consequently, the immediate priority is not the development of additional policies, but the operationalization of existing governance instruments before progressively expanding the governance framework into additional domains.

The three development phases are summarized below.

Phase I. Operationalization of the Data Management and Protection Policy

This phase focuses on translating the approved Data Management and Protection Policy into operational practice. Emphasis will be placed on identifying the SOPs required under each policy article, developing priority procedures supporting the DPO Office, and producing the forms, templates, notices and institutional registers necessary to facilitate implementation.

Phase II. Expansion of the IDGF

Following successful operationalization of the existing policy, the Framework will be expanded through the progressive development of additional governance policies covering priority institutional governance domains identified in PART I. Each policy will be accompanied by the operational and technical documentation required to support implementation.

Phase III. Institutionalization of the Framework

The final phase focuses on integrating all governance instruments into a single institutional governance framework. Activities will include institutional roll-out, governance coordination, capacity building, monitoring and evaluation, continual improvement, and periodic review to ensure long-term sustainability of the Framework.

The outputs of each phase will collectively contribute towards the establishment of a comprehensive IDGF that supports effective governance of institutional data assets throughout their life-cycle while strengthening research excellence, institutional accountability, regulatory compliance and evidence-informed decision-making.

2.12 Phase I: Operationalizing the Data Management and Protection Policy

This phase focuses on developing the governance instruments required to operationalize the approved Data Management and Protection Policy across the Institute.

2.12.1 Objectives

The objectives of Phase I are to:

- a) Assess the Institute's current level of compliance with the DPA and the approved Data Management and Protection Policy.
- b) Develop the institutional Data Protection Compliance Implementation Strategy to guide the phased implementation of data protection governance across the Institute.
- c) Establish an institutional network of Data Protection Champions to coordinate implementation, promote awareness and support compliance activities within Directorates, Departments and Sections.
- d) Develop priority Standard Operating Procedures required for the effective operationalization of the Data Management and Protection Policy.
- e) Develop the supporting forms, templates, notices, agreements, registers and other controlled documents required to implement the approved Standard Operating Procedures.
- f) Develop a costed implementation work plan to support the phased implementation of approved governance instruments.
- g) Strengthen institutional compliance with the DPA, associated Regulations and the approved Data Management and Protection Policy.

2.12.2 Data protection compliance implementation strategy

The successful operationalization of the Data Management and Protection Policy will require a coordinated institutional implementation strategy that extends beyond the Office of DPO. Effective implementation will depend on active participation by all Directorates, Departments and Sections responsible for generating, processing, storing or managing institutional data.

Accordingly, this phase (phase I) will establish an institutional data protection compliance implementation strategy that defines governance responsibilities, implementation priorities, reporting arrangements and institutional coordination mechanisms. The Office of DPO will provide overall technical leadership and regulatory oversight, while Heads of Directorates, Departments and Sections will be responsible for implementing approved governance instruments within their respective functional areas.

To facilitate institution-wide implementation, each Division will nominate a Data Protection Champion to serve as the focal person for coordinating compliance activities, supporting awareness programs, monitoring implementation, and liaising with the Office of the DPO. The Champions will provide an institutional coordination mechanism that promotes consistent implementation of the Policy while strengthening communication between operational units and the DPO Office.

The implementation strategy will also establish institutional reporting and monitoring arrangements to support progressive implementation of approved governance instruments, enabling KIPRE to monitor compliance, identify implementation challenges and continuously improve institutional data protection practices.

2.12.3 SOPs inventory

The successful implementation of the Data Management and Protection Policy requires documented SOPs that translate the Policy provisions into consistent institutional practice.

Accordingly, each substantive article of the Policy was systematically reviewed to identify the operational procedures necessary to support its implementation.

This review identified thirteen SOPs that collectively constitute the operational framework required to implement the current Data Management and Protection Policy. The proposed SOP inventory is presented in Table 3, together with the lead office responsible for each procedure, the corresponding policy provisions (articles) and the proposed development priority.

Recognizing that the development of all SOPs cannot be undertaken simultaneously, implementation will be phased according to institutional priorities, regulatory obligations and operational needs. SOPs assigned a **High** development priority constitute the initial implementation package to be developed during Phase I, while those assigned a **Medium** priority will be progressively developed as implementation of the Policy matures. Additional SOPs required to operationalize future governance policies will be developed during the subsequent expansion of the IDGF.

Proposed SOP	Lead office	Policy article(s)	Primary purpose	Dev. priority
SOP-001: Joint Ethical and Privacy Compliance Review	DPO Office / ISREC	Articles 2–4	Integrate privacy vetting rules directly into active human and animal research ethical clearance loops before project lifecycle deployment to eliminate duplicate administrative loops.	High
SOP-002: Data Subject Rights and Research Defenses	DPO Office	Articles 3–4	Operationalize corporate data rights requests (access, correction) while enforcing strict verification to block premature deletion of clinical research records.	High
SOP-003: Data Protection Impact Assessment (DPIA)	DPO Office	Articles 2, 3 & 10	Establish procedures for conducting pre-screening Data Protection Impact Assessments for high-risk research algorithms and clinical trials.	High
SOP-004: Personal Data Lifecycle Management	DPO Office	Article 4	Standardize lawful collection, storage parameters, and access controls for personal data across corporate, HR, and facility visitor registries.	High
SOP-005: Research Data Stewardship	RPDD / DSAS	Article 4	Deploy mandatory pre-project Data Management Plans (DMPs) and stewardship agreements for Principal Investigators during Phase I.	High
SOP-006: Data Sharing and Disclosure	DPO Office / Legal Office	Article 5	Govern internal and international sharing of data registries, ensuring strict enforcement of Data Use Agreements (DUAs) and cross-border safeguards.	High

Proposed SOP	Lead office	Policy article(s)	Primary purpose	Dev. priority
SOP-007: Information Security and Data Protection Controls	ICT Division	Article 6	Define minimum technical, organizational and physical safeguards for protecting institutional information assets and personal data.	Medium
SOP-008: Personal Data Breach and 4-Hour Escalation	DPO Office / ICT Division	Article 7	Standardize internal technical failure logging within 4 working hours and manage containment, triage, and statutory 72-hour notifications to the ODPC.	High
SOP-009: Data Retention, Archiving and Secure Disposal	DPO Office / Registry / ICT	Article 8	Standardize retention schedules, archival procedures and secure disposal of institutional data and records.	Medium
SOP-010: Data Protection Complaints and Dispute Resolution	DPO Office	Article 9	Establish procedures for receiving, investigating and resolving complaints relating to personal data processing.	Medium
SOP-011: Data Protection Monitoring, Audit and Compliance	DPO Office	Articles 10–12	Operationalize compliance monitoring, internal audits, reporting and continual improvement of institutional data protection practices.	Medium
SOP-012: Third-Party Data Processing and Contract Management	Legal and DPO Offices	Articles 3 & 5	Govern the engagement, monitoring and oversight of third-party processors and contractual data protection obligations.	Medium
SOP-013: Data Protection Awareness and Training	CBPGMD & DPO Office	Articles 2 & 6	Standardize institutional awareness, induction and periodic data protection training for staff, students and collaborators.	Medium

Proposed SOP	Lead office	Policy article(s)	Primary purpose	Dev. priority
--------------	-------------	-------------------	-----------------	---------------

Table 3: Proposed SOPs required to operationalize the Data Management and Protection Policy.

2.12.4 Priority SOPs for the DPO Office

Although the operationalization of the Data Management and Protection Policy requires the progressive development of all identified SOPs, the initial implementation effort will focus on those that directly support the statutory and operational responsibilities of the DPO office. These procedures establish the institutional mechanisms required to implement the DPA and provide the operational foundation upon which the broader IDGF will be progressively developed.

The priority SOPs have been selected based on three considerations. First, they address the statutory obligations imposed on the Institute as a Data Controller and Data Processor under the DPA. Second, they establish the core institutional processes required for lawful processing of personal data across research, surveillance, consultancy and administrative activities. Third, they provide the governance infrastructure upon which the remaining institutional SOPs can be implemented in a coordinated and consistent manner.

The priority SOPs proposed for immediate development are presented in Table 4.

Priority SOP	Primary justification
Joint Ethical and Privacy Compliance Review	Provides the institutional gatekeeper mechanism integrated directly into active human and animal research ethical clearance loops (AREC/HREC) to streamline project approvals without duplicate administrative paths.
Data Subject Rights and Research Defenses	Establishes clear operational tracks for resolving corporate data access requests while enforcing strict data retention defenses to safeguard ongoing clinical trials and disease surveillance metadata from premature erasure.
Data Protection Impact Assessment (DPIA)	Provides a standardized pre-screening risk methodology for assessing privacy risks and algorithmic accountability associated with high-risk biomedical processing activities and automated health profiling projects.
Personal Data Lifecycle Management	Standardizes the lawful collection, processing boundaries, security tracking, and automated deletion milestones for personal data handled within human resources, procurement, and facility visitor registries.
Data Sharing and Disclosure	Establishes institutional procedures and contractual vetting standards governing the lawful sharing of personal data within internal KIPRE divisions and with external international research collaborators.
Personal Data Breach and 4-Hour Escalation	Provides structured protocols for technical compromise containment and triage, explicitly embedding the internal 4 working hour system failure reporting window and the statutory 72-hour notification to the ODPC.

Table 4: Priority SOPs for the DPO Office.

Collectively, these six (6) SOPs constitute the minimum operational package required to establish the DPO Office as the institutional focal point for privacy governance and regulatory compliance. Their implementation will provide the immediate operational capability required to support institutional compliance while creating the governance foundation upon which the remaining SOPs identified in Table 3 will be progressively developed.

2.12.5 Supporting implementation documents

The implementation of the priority Standard Operating Procedures requires a suite of controlled implementation documents that support consistent execution, documentation, communication and institutional accountability. These documents provide the operational tools necessary for implementing approved procedures, recording institutional decisions, demonstrating compliance and maintaining appropriate governance records.

The implementation documents developed during Phase I will primarily support the operational functions of the Office of the Data Protection Officer. As additional governance policies are developed under subsequent phases of the IDGF, corresponding implementation documents will be developed to support those governance instruments.

Priority SOP	Proposed implementation documents	Responsible office
Joint Ethical and Privacy Compliance Review	• Data Protection Compliance Review Form • ISREC Pre-Screening Compliance Checklist • Institutional Project Clearance Register • Privacy Compliance Clearance Letter	Office of the DPO / ISREC Secretariat
Data Protection Impact Assessment (DPIA)	• DPIA High-Risk Screening Questionnaire • Algorithmic Risk Assessment Template • Biomedical Project DPIA Report Template • Institutional Central DPIA Register	Office of the DPO
Data Subject Rights and Research Defenses	• Data Subject Rights Application Form (DPO-01A) • Applicant Identity Verification Protocol • Statutory Data Disclosure Letter Template • Central DSAR Compliance Register (DPO-01B)	Office of the DPO
Personal Data Lifecycle Management	• Institutional Intranet Privacy Notice Templates • Institutional Consent Form Templates (Annex III) • Withdrawal of Personal Consent Form • Corporate Service Privacy Statements	Office of the DPO / Corporate Services
Research Data Stewardship	• Research Data Management Plan (DMP) Template • LIMS Specimen Metadata Template • Principal Investigator Data Custody Agreement Form • Central Integrated Research Inventory Register	RPDD / DSAS

Priority SOP	Proposed implementation documents	Responsible office
Data Sharing and Disclosure	• External Data Sharing Request Form • Multi-Institutional Data Sharing Agreement (DSA) • Core Data Processing Agreement (DPA) • Non-Disclosure Agreement Template (Annex I) • International Data Use & Transfer Agreement (DTA) • Centralized Data Disclosure Register	Legal Office / Office of the DPO
Personal Data Breach and 4-Hour Escalation	• Internal Security Incident Ingestion Form (DPO-02A) • ICT Division 4-Hour Incident Log Framework • Statutory 72-Hour ODPC Breach Notification Template • Personal Data Breach Ledger	Office of the DPO / ICT Division
Information Security and Data Protection Controls	• User Access Privilege Request Form • Role-Based Access Control (RBAC) Register • Institutional ICT Network Asset Register • Secure Cryptographic Storage Disposal Certificate	ICT Division / Office of the DPO

Table 5: Proposed supporting implementation documents for the Phase I priority Standard Operating Procedures.

2.12.6 Expected outputs

Implementation of Phase I is expected to produce the following outputs:

- a) A fully operational implementation framework supporting the Data Management and Protection Policy.
- b) Priority SOPs for the Office of the DPO developed and approved.
- c) Standardized forms, templates, notices and institutional registers supporting implementation of the priority SOPs.
- d) Standardized institutional procedures for compliance review, personal data management, data sharing, data subject rights management, data breach management and Data Protection Impact Assessments.
- e) Improved institutional compliance with the DPA, the Data Protection Regulations and the KIPRE Data Management and Protection Policy.
- f) A governance foundation for the progressive expansion of the IDGF.

2.13 Phase II: Expansion of the governance framework

Having established the operational foundation for the Data Management and Protection Policy under Phase I, the next stage focuses on the development of the broader IDGF. This phase aims to expand institutional data governance beyond data protection by developing the additional governance instruments required to support the effective management, stewardship and utilization of institutional data assets.

Given the multidisciplinary and highly technical nature of institutional data governance, this phase will involve the engagement of a technical consultant to work collaboratively with KIPRE in designing and developing the remaining components of the governance framework. The consultant will work closely with the Office of the DPO, the DSAS, the ICT Division, the RPDD, and other relevant stakeholders to ensure that the Framework aligns with national legislation, international best practices, and the strategic objectives of the Institute.

2.13.1 Purpose of Phase II

The purpose of Phase II is to progressively develop the remaining components of the IDGF by establishing additional governance policies together with their supporting implementation instruments. Collectively, these governance instruments will provide the institutional architecture required for governing research data, institutional information assets, data quality metrics, advanced analytics, and emerging artificial intelligence platforms across their entire operational lifecycle.

2.13.2 Governance policies to be developed

The proposed governance policies and technical standard frameworks to be developed under Phase II are presented in Table 6 and are also outlined in Table 2 of Part I.

Governance Policy / Framework Addendum	Primary Focus and Operational Delivery Mode
Artificial Intelligence (AI) Governance Policy	Establish strategic principles for the responsible acquisition, validation, deployment, and oversight of Machine Learning and predictive models tracking emerging zoonotic or preclinical health trends.
Data Quality Standard Addendum (KIPRE ICT Policy Addendum)	Enforce precise engineering validation rules and database schema entry criteria across the LIMS platform in strict alignment with international ISO 8000 Data Integrity baselines.
Research Data Lifecycle Guidelines (KIPRE Research Policy Addendum)	Operationalize long-term scientific data preservation, stewardship, and structured metadata repository tracking to guarantee reproducibility across all directorates.
Open Science and FAIR Data Sharing Guidelines	Govern the public and international sharing of data registries, ensuring strict enforcement of Data Use Agreements (DUAs) while optimizing collaborative visibility.

Table 6: Proposed governance policies and framework additions for development under Phase II.

2.13.3 Supporting governance instruments

Each governance policy and framework addendum developed under Phase II will be accompanied by a suite of controlled implementation documents necessary to support consistent institutional

application. These governance instruments will include:

- Standard Operating Procedures (SOPs);
- Technical Standards;
- Operational Guidelines;
- Intake and Compliance Forms;
- System Integration Templates;
- Privacy and Processing Notices;
- Audit Checklists; and
- Centralized Governance Registers.

The specific implementation instruments required for each governance policy will be systematically identified during the strategic policy development process and incorporated into the master IDGF architecture.

2.13.4 Governance structures and stewardship

Successful implementation of the IDGF requires clearly defined governance responsibilities across the Institute. Table 7 presents the proposed governance structure for institutional data stewardship.

Governance function	Lead responsibility
Office of the DPO	Privacy governance, regulatory compliance oversight, and external liaison with the Office of the Data Protection Commissioner (ODPC).
DSAS	Core institutional data governance, data stewardship tracking, data quality framework architecture, and analytics governance.
ICT Division	Information security, technical network infrastructure, server administration, user access privilege management, and cybersecurity controls.
RPDD	Governance of research data datasets, scientific integrity monitoring, and institutional research compliance.
Directorate Heads	Implementation of approved data governance requirements and policy directives within their respective functional directorates.
Data Stewards	Operational oversight of localized institutional datasets, metadata curation, and data quality metrics within assigned functional areas.
Data Custodians	Technical management, safe storage architecture, and backend cryptographic protection of institutional data assets under delegated authority.

Table 7: Proposed governance structure and stewardship responsibilities under the IDGF.

2.14 Phase III: Institutionalization of the IDGF

Following the strategic development of the expanded policy portfolio and supporting implementation instruments under Phase II, the final stage focuses on the structural institutionalization of the completed IDGF across KIPRE. The overarching objective of this phase is to seamlessly embed data governance principles into routine institutional operations, ensuring that data integrity, statutory privacy, and asset optimization become native to the organizational culture, executive decision-making, and day-to-day laboratory and corporate activities.

True institutionalization extends far beyond the mere publication of controlled governance text. It demands a sustained, top-down organizational commitment, explicitly mapped data stewardship responsibilities, continuous human resource capacity development, proactive compliance monitoring, and the iterative refinement of governance instruments in response to shifting technological, legislative, and international scientific research parameters.

2.14.1 Framework integration

The completed IDGF will be systematically engineered into KIPRE's existing research administration, legal vetting, corporate support, and ICT infrastructural operations. Privacy-by-design and data quality validation gates will be hardcoded directly into the initial project lifecycle clearance workflows managed by the ISREC Secretariat, the DRPD project intake channels, and the procurement vetting cycles handled by the Legal Office. By embedding strict governance checkpoints within the core Laboratory Information Management Systems (LIMS), enterprise resource planning networks, and international partnership frameworks, the Institute ensures that data governance functions as a mandatory, automated catalyst for operational excellence rather than an isolated administrative hurdle.

2.14.2 Capacity building and cultural sensitization

Achieving enterprise-wide compliance necessitates a robust, continuous capability-building program executed across all directorates, divisions, and research stations. To achieve this, the DPO Office, in coordination with the Directorate of Capacity Building, Partnerships, and Grants Management (CBPGMD), will operationalize a formalized training syllabus. This curriculum will deliver targeted data privacy and data handling orientation courses tailored for standard staff, visiting researchers, post-graduate students, and external collaborators.

Crucially, this phase will activate and empower the nominated network of **Data Protection Champions** within each technical division. These Champions will serve as localized domain experts, spearheading peer-to-peer compliance reinforcement, identifying localized security gaps, and serving as the direct operational loop connecting frontline laboratory technicians and administrative officers with the central DPO Office.

2.14.3 Monitoring, evaluation, and compliance assurance

The progressive implementation of the IDGF will be subjected to strict, data-driven monitoring and evaluation (M&E) mechanisms to verify control adequacy and ensure statutory accountability. Compliance assurance will be driven by independent risk assessments executed **biannually** by the Internal Audit and Risk Assurance Division, supplemented by regular DPO compliance spot-checks

and automated network system audits.

To transition this oversight from a qualitative checklist into an empirical evaluation matrix, KIPRE will track a standardized set of key performance indicators (KPIs), which include:

- The percentage of active research projects maintaining fully audited Data Management Plans (DMPs);
- System uptime and mean-time-to-resolution (MTTR) for infrastructure failures within the mandatory 4-hour DPO ingestion window;
- The volume and resolution speed of Data Subject Access Requests (DSARs) processed strictly within the 30-day statutory window;
- Staff training completion rates across high-exposure registries (HR, Finance, LIMS Nodes); and
- The frequency and closure velocity of internal data protection compliance review gaps.

2.14.4 Continuous improvement and evolutionary governance

The IDGF is engineered as a dynamic, evolutionary ecosystem designed to scale alongside shifting regulatory environments and technological innovations. In accordance with institutional data maturity requirements, all governance policies, cross-functional SOPs, technical standards, and controlled implementation forms will be subjected to a mandatory, systematic review every three (3) years, or sooner if necessitated by significant national legislative amendments or technological paradigm shifts.

Forensic audit findings, corporate risk ledger entries, collaborative researcher feedback, and breakthroughs in advanced data analytics or secure cryptographic erasure methodologies will be continuously analyzed. This iterative loop guarantees that KIPRE maintains a highly sophisticated, adaptive, and defensive data governance program capable of securing its status as a global biomedical center of excellence.

3 Overall implementation road-map

The implementation of the IDGF will be undertaken progressively through three complementary phases. Each phase systematically builds upon the achievements and structural milestones of the preceding phase to ensure that KIPRE's institutional data governance capabilities mature in a structured, coordinated, and sustainable manner (see Table 8 below).

4 Conclusion

The conceptualization and formulation of the Institutional Data Governance Framework (IDGF) represents a landmark milestone toward modernizing the security, administration, and optimization of data assets across the Kenya Institute of Primate Research (KIPRE). By embracing a structured, phased implementation roadmap, the Institute moves beyond baseline regulatory checkboxes to systematically build the technical policies, operational standard operating procedures, data quality standards, and controlled tools necessary to secure deep institutional data maturity.

Phase	Major Activities	Lead Responsibility	Expected Strategic Outcome
Phase I	Operationalize the Data Management and Protection Policy through the deployment of high-priority SOPs, specialized intake forms, and centralized compliance registers.	Office of the Data Protection Officer (DPO)	A fully functional, audit-ready data protection infrastructure and operational compliance framework aligned to ODPC standards.
Phase II	Expand the target governance portfolio by drafting standalone policies, framework addenda, technical standards, and operational workflows via specialized external consultancy.	Technical Consultant in collaboration with DSAS, ICT, and Legal	Comprehensive Institutional Data Governance Framework (IDGF) established and cross-referenced with KIPRE's core research mandate.
Phase III	Institutionalize the completed framework through cross-functional integration, continuous capacity building, Data Protection Champions activation, and strict KPI monitoring.	KIPRE Senior Management and Directorate Heads	A sustainable, adaptive, and enterprise-wide data governance program that transforms data assets into strategically secured institutional resources.

Table 8: Overall implementation road-map for the IDGF.

The successful rollout of this Development Plan not only translates the active Data Management and Protection Policy (POL/KIPRE/DRPD/002) into a living laboratory and administrative reality, but it also delivers the overarching governance architecture necessary to sustain research excellence, eliminate information fraud risks, enforce organizational accountability, and drive responsible health technology innovation. Through continuous implementation, independent compliance auditing, and recursive improvement cycles, the completed IDGF will function as a highly resilient, enterprise-wide asset protection paradigm that shields KIPRE's proprietary data and advances its long-term mandate as a global biomedical center of excellence.

5 References

- [1] Constitution of Kenya. (2010). Government of Kenya.
- [2] Data Protection Act, No. 24 of 2019. Government of Kenya.
- [3] Data Protection (General) Regulations, 2021. Government of Kenya.
- [4] Computer Misuse and Cybercrimes Act, No. 5 of 2018. Government of Kenya.
- [5] Access to Information Act, No. 31 of 2016. Government of Kenya.
- [6] Kenya Information and Communications Act (Cap. 411A). Government of Kenya.
- [7] Public Archives and Documentation Service Act (Cap. 19). Government of Kenya.
- [8] Ministry of Information, Communications and the Digital Economy. (2022). Kenya National Digital Master Plan 2022–2032.
- [9] Ministry of Information, Communications and the Digital Economy. (2025). Kenya National Artificial Intelligence Strategy 2025–2030.
- [10] DAMA International. (2017). *The DAMA Guide to the Data Management Body of Knowledge (DAMA-DMBOK2)* (2nd ed.). Technics Publications.
- [11] Organisation for Economic Co-operation and Development (OECD). (2023). *Data Governance in the Public Sector*. OECD Publishing.
- [12] UNESCO. (2021). *Recommendation on the Ethics of Artificial Intelligence*. United Nations Educational, Scientific and Cultural Organization.
- [13] International Organization for Standardization (ISO). (2017). *ISO/IEC 38505-1: Information technology — Governance of IT — Governance of data*.
- [14] International Organization for Standardization (ISO). (2022). *ISO/IEC 27001: Information security, cybersecurity and privacy protection — Information security management systems — Requirements*.
- [15] International Organization for Standardization (ISO). (2019). *ISO/IEC 27701: Privacy Information Management Systems*.
- [16] Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3, 160018.
- [17] UNESCO. (2021). *UNESCO Recommendation on Open Science*. United Nations Educational, Scientific and Cultural Organization.
- [18] Kenya Institute of Primate Research (KIPRE). (2023). *Strategic Plan 2023–2027*.
- [19] Kenya Institute of Primate Research (KIPRE). (2026). *Data Management and Protection Policy*.
- [20] Kenya Institute of Primate Research (KIPRE). (2026). *Institutional Data Management Needs Assessment Report*.
- [21] Office of the Data Protection Commissioner (ODPC). (2026). *Certificate of Registration as Data Controller and Data Processor*.